

Primary source peer-reviewed scholarly articles frequently use vocabulary unfamiliar to the average reader. Computer-generated summaries will often include these terms without defining them.

- **Important nutrients:** The study highlights the importance of foods such as fruits and vegetables, which are rich in nutrients such as folate, magnesium, b-group vitamins, selenium, zinc, mono- and polyunsaturated fatty acids, polyphenols and fibre that have been reported as being relevant in depressive illnesses. These nutrients are important **modulators of redox status** and immune system functioning, which are both **pathophysiological substrates of depressive illness**.
- **Dietary Shifts:** The study notes that there has been a reduction in the quality of adolescents' diets in recent decades, with a reduction in consumption of raw fruits, high-nutrient vegetables and dairy foods, and an increase in consumption of fast food, snacks and sweetened beverages.
- **Inflammation:** The study notes the important role of inflammation in the development of depression.
 - Consumption of a diet rich in antioxidants, vitamins, minerals and fiber is associated with reduced systemic inflammation.
 - Diets that are low in essential nutrients and western type dietary patterns are associated with increased systemic inflammation.
 - Hyperglycemia promotes an inflammatory state, and **high glycemic load diets** are also associated with increased systemic inflammation.
- **Neurotrophic Factors:** The study suggests that dietary factors, such as refined sugars and saturated fats, have a detrimental impact on the expression of neurotrophic factors that are particularly salient to depressive illness.

Select one of the four highlighted terms and provide the following information in your answer:

1. A definition you found online
2. Cite the source of the definition
3. Write an original sentence demonstrating your understanding of the term

Your Answer:

Modulators of redox status

1. A definition for redox: "A chemical reaction that takes place between an oxidizing substance and a reducing substance. The oxidizing substance loses electrons in the reaction, and the reducing substance gains electrons." (National Cancer Institute, n.d.)
2. National Cancer Institute. (n.d.). redox. <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/redox>
3. Redox is a chemical reaction between two substances where the oxidizing substance loses electrons, and the reducing substance gains electrons. Now that I understand the term redox, I can imply that its status would be the state at which the two substances are at. I.e., How many electrons they each have. Then the modulation of that redox status is the way it is being balanced; not too many electrons being given or taken away. And since those nutrients listed (folate, magnesium, b-group vitamins, selenium, zinc, mono- and polyunsaturated fatty acids, polyphenols, and fiber.) are modulators of the redox status, they are the ones keeping the redox reaction in check.